GEOX — State of the Organ

Compiled brief · 2026-10-20 02:30 MYT · for Arif Fazil (F13)

VERDICT IN ONE PARAGRAPH

GEOX is **running, deployed, and materially further ahead than at my last freeze** — two new commits from another agent added a deterministic evidence-to-decision chain, and the service is live on `a93747b1`. Three things remain true: the **public MCP edge still returns 502 to a non-conforming client**, the **Kinabalu Moho chain rests on a citation whose DOI resolves to a different paper**, and **scientific readiness is still zero** — no canonical transect, no GEBCO/TID asset, no cross-section derived from registered objects. Governance is intact; the science is not yet.

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1. LIVE STATE (all OBS, probed this session)

Field	Value
Service	`geox-mcp` active, 0 restarts
Deployed SHA	`geox-a93747b1`
Repo HEAD	`a93747b1` · dirty 2 (`apps/index.html`, `scripts/publish_docroot.py`)
Tools / resources / prompts	**27 / 52 / 16**
Surface drift	`ok = true` (canonical 27 = live 27)
Deployment drift	`drift = true`
Kernel verdict inherited	**HOLD** — from arifOS, not a GEOX fault
`/health` (public)	200
`/mcp` no `Accept` header (public)	**502** — framing defect still live
P0 framing patch	**not applied** (`_rebuild` occurrences = 0)

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2. WHAT OTHER AGENTS SHIPPED SINCE `778dcb58`

GEOX — 2 commits, 1,215 insertions

- `0c62f150` — *deterministic evidence-to-decision chain + physics section output path fix* New: `src/geox_core/evidence_chain.py` (216 lines), `schemas/earth/evidence_chain.schema.json` (264), `scripts/run_evidence_chain.py` (157), `tests/test_evidence_chain.py` (164), and a worked artifact `data/evidence/kinabalu_moho_chain.json`.
- `a93747b1` — *truth-up surfaces from registry.py with 0-drift gate* (surface no longer hand-typed).

arifOS — `a4013ab5d` by agy/FI-004 *fix(crypto_auth): align 888 and opencode private key paths and loopback auto-sign* — this commits the auth-map correction I applied and proposed. Two of my three items are now in the kernel; item 2 (`FI-008` key) is still unmatched.

Verified mechanically: `pytest tests/test_evidence_chain.py` → **6 passed in 0.13s**.

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3. THE KINABALU MOHO CHAIN — AND ITS AUDIT

What it claims (OBS from the artifact): Moho depth **32.8 ± 2.5 km** at station `KMY_BROADBAND` (5.98°N, 116.07°E), from P-to-S receiver-function H- stacking (Zhu & Kanamori 2000). Discrimination run: **Model A (underthrust) PASS**, residual 0.0 · **Model D (lithospheric drip) FAIL**, residual 9.2 km vs threshold 2.5 km. `confidence_score 0.88`.

Audit (four existence checks, `evidence-chain-audit`):

Check	Result	Verdict
Citation resolves	DOI `10.7186/bgsm62201607` → **"CHIRP acoustic characterization of paleo fluvial system of Late-Pleistocene..."**, not "Crustal thickness beneath North Borneo…" (Sidek et al.)	**EXISTS_WRONG_ENTITY**
Citation title search	Crossref bibliographic query returned no match for the Sidek paper	**UNCHECKED**
Hash repetition	`file_hash`, both `geometry_hash`, `artifact_identity` — **no repeating unit**	shape-valid
Hash referent	no file on disk hashes to `file_hash` (paper not in repo)	**UNCHECKED**
Witness environment	`Linux 6.17.0-41-generic x86_64` matches `uname`	environment OK
Witness PID	`961252` no longer exists (service restarted since)	inconclusive
Witness commit	run stamped `geox-778dcb58`, repo now `a93747b1`	**stale**
Test run	6/6 pass under the venv serving the service	**EXISTS_MATCHES**

Two corrections I owe on my own work in this session: 1. I first reported the hashes as `FABRICATED_SHAPE`. **That was a false positive** — my test counted unique hex characters, and hex has only 16. Retracted. 2. `geometry_hash` is **25 characters**, not 64. That is a *labelling* defect (truncated digest presented as a hash), not fabrication.

Net: the discrimination is arithmetically sound and test-covered; its **load-bearing input is miscited**. Until the DOI is corrected or the paper's true identifier is found, `confidence_score 0.88` is unsupported at the top and the `observed:` tier should read `UNVERIFIED_CITATION`.

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4. WHAT IS ACTUALLY BROKEN

#	Defect	Severity	Owner
1	`_CacheableListResponseMiddleware` drains `body_iterator` then returns the drained response → 0-byte body → edge **502**	**P0** — breaks every JSON/gzip client	GEOX
2	Evidence-chain DOI points at an unrelated paper	**P0 for the science** — poisons the chain's top	GEOX
3	`TOOL_TIMEOUTS` declared, never referenced	P1	GEOX
4	SEP-1303: argument errors arrive as protocol −32001, not `isError`	P1	GEOX
5	Anonymous rate limit keyed on a per-call session → unenforceable	P1	GEOX
6	2 of 13 MCP Apps inert (`basin-explorer`, `catalog`)	P2	GEOX
7	`geometry_hash` 25 chars presented as a hash	P2	GEOX
8	deploy lane dirty from a sibling's in-flight work	ops	—

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5. WHAT IS NEXT — ranked

Tier 1 — restore client trust (days) 1. Apply the framing patch (`abd0bcca…`, already validated 12/12 mechanism) to `server.py`. Smallest demonstrated cause: two `return response` sites after a drained iterator. 2. Add an existence gate to the evidence-chain pipeline: DOI must resolve **and** title/authors must match; hash must be recomputed from a retrieved artifact.

Tier 2 — make the chain honest (weeks) 3. Correct or replace the Sidek et al. reference; re-run the discrimination; restamp `observed` with the true DOI and the paper's SHA-256. 4. Re-run the chain at the current commit so the witness is not stale.

Tier 3 — the actual mission (the real work) 5. Canonical transect object: ordered coordinates, explicit CRS, chainage, vertical datum, source hash. 6. Bounded GEBCO subset **plus matching**

TID subset — categorical, never averaged; NoData explicit. 7. Map and cross-section derived from that one object — so the Kinabalu page stops being hand-placed geometry.

Not done, and should not be claimed: no canonical transect, no GEBCO/TID asset, no cross-section generated from registered objects, no vertical-datum validation, no coastal seam test.

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6. PROVENANCE

compiled_at	2026-10-20T02:30 MYT
geox_repo	a93747b1 (dirty 2)
geox_live	geox-a93747b1
kernel	a4013ab5d
evidence_artifact	data/evidence/kinabalu_moho_chain.json (6,488 B)
tests	6 passed (tests/test_evidence_chain.py)
P0 patch	/tmp/opencode/mission/geox-p0-1.patch sha256 abd0bcca… UNAPPLIED
claims	labelled OBS / DER / INT / SPEC inline

No deploy, restart, seal, or ledger write was performed in compiling this brief.